

Jeehong Kim

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RESEARCH INTEREST

I am interested in designing broadly applicable machine learning techniques for graph-based problems, especially in heterophilous and anomalous scenarios.

- **Keywords:** Graph Representation Learning, Graph Heterophily, Anomaly Detection, GNNs with LLM

EDUCATION

Seoul National University, Seoul, South Korea

- M.S.-Ph.D. Integrated Program in Data Science Mar 2023 – Present
 - Research Interest: Graph Representation Learning, Graph Heterophily, Anomaly Detection
 - Advisor: Prof. Hyunwoo Park
- Planning to fulfill military service as a *Technical Research Personnel* starting around September 2026

Korea University, Seoul, South Korea

- B.S. in Industrial Management Engineering Mar 2019 – Feb 2023
 - GPA: 3.76 / 4.50

RESEARCH EXPERIENCE

DSAIL (Data Science & Artificial Intelligence), KAIST

- Undergraduate Research Intern Jun 2022 – Aug 2022
 - Research Interest: Graph Neural Network, Recommendation System
 - Advisor: Prof. Chanyoung Park

AIDA Lab (Artificial Intelligence & Data Analytics), Korea University

- Undergraduate Research Intern May 2021 – Jan 2022
 - Research Interest: Anomaly Detection, GAN
 - Advisor: Prof. Sungwon Han

PUBLICATIONS

(†: Equal contribution)

JOURNALS

- [J2] STGVAD: Spatio-Temporal Graph-based Vessel Behavior Anomaly Detection
Jeehong Kim[†], Minchan Kim[†], Youngseok Hwang, Sungho Bae, Deuk Jae Cho, Wonhee Lee, Hyunwoo Park
IEEE Access (2025.09)
- [J1] FURIOUS: Fully unified risk-assessment with interactive operational user system for vessels
Yooyeun Kim[†], **Jeehong Kim**[†], Wonhee Lee, Hyunwoo Park, Deuk Jae Cho
PLoS One (2025.05)

WORKSHOPS

- [W2] Spatio-Temporal Graphs Beyond Grids: Benchmark for Maritime Anomaly Detection
Jeehong Kim[†], Youngseok Hwang[†], Minchan Kim, Sungho Bae, Hyunwoo Park
NeurIPS 2025 Workshop in AI for Science: The Reach and Limits of AI for Scientific Discovery
- [W1] Adaptive Sparsified Graph Learning Framework for Vessel Behavior Anomalies
Jeehong Kim[†], Minchan Kim[†], Jaeseong Ju, Youngseok Hwang, Wonhee Lee, Hyunwoo Park
AAAI 2025 Workshop in Anomaly Detection in Scientific Domains

PROJECTS

MOIN

- Fintech Technology Commercialization Support Program Jul 2025 – Present
 - Fraud detection system (FDS) modeling optimization for overseas remittance using AI technology

Korea Research Institute of Ships & Ocean Engineering (KRISO)

- Vessel Behavior Anomaly Detection Oct 2024 – Present
 - Anomaly detection using a GNN-based framework on a customized dataset
 - (Published) *Adaptive Sparsified Graph Learning Framework for Vessel Behavior Anomalies*
 - (Published) *STGVAD: Spatio-Temporal Graph-based Vessel Behavior Anomaly Detection*
- Visualization for Ship Collision Risk Assessment Mar 2024 – May 2025
 - Development of a decision supportive interactive system for ship collision avoidance
 - (Published) *FURIOUS: Fully unified risk-assessment with interactive operational user system for vessels*

VAIV Company

- Generative AI Professional Development Project Mar 2024 – Present
 - Task planning with LLM-guided GNNs

	Seoul National University ▪ SNU Creative Education Project (Inno-Edu 2031) Jan 2024 – Dec 2024 • Curriculum reform initiative under the SNU Creative Education Project ▪ Undergraduate Course Development Jan 2024 – Jun 2024 • (L0655.005100) Veritas Practice: Data Visualization and Me
AWARDS & SCHOLARSHIPS	Best Poster Award Dec 2025 ▪ NeurIPS 2025 Workshop in AI for Science: The Reach and Limits of AI for Scientific Discovery BK 21 Scholarship , Brain Korea Mar 2025 – Present SNU Medical AI Scholarship , Seoul National University Aug 2024 – Present Veritas Scholarship , Korea University Feb 2022 ▪ Research on detection of anomalies in the manufacturing process • Advisor: Prof. Sungwon Han HAICON2021 , Korea Institute of Information Security & Cryptology (KIISC) Nov 2021 ▪ Construction of an AI model for identifying time-series based anomalies in industrial control systems • Ranked 6th out of 177 teams Industrial Engineering Project Competition , Korean Institute of Industrial Engineers (KIIE) Oct 2021 ▪ Design of a dynamic corridor network for K-UAM • Received participation award in industrial engineering project competition Data Visualization Competition , Korea University, Yonsei University, Intel Aug 2021 ▪ Visualization of patient and their deductible premium relationship • Received grand prize
PROFESSIONAL SERVICES	Conference Review ▪ Learning on Graphs Conference (LOG) 2025 Workshop Review ▪ The Reach and Limits of AI for Scientific Discovery (AI4Science)@NeurIPS 2025 2025 Journal Review ▪ IEEE Access 2025
ACTIVITIES	Lab Captain , Visualization and Business Analytics Lab (ViBA) Dec 2024 – Present ▪ Advisor: Prof. Hyunwoo Park Samsung Dream Class , Samsung Welfare Foundation Sep 2021 – Feb 2022 Korea-China Leadership Program , Korea Foundation for Advanced Studies (KFAS) Jul 2019
TEACHING EXPERIENCE	Teaching Assistant ▪ K-Digital training program at Seoul National University Jun 2024 ▪ Python data analysis training program for Seoul National University faculty members Jan 2024
TALKS AND SEMINARS	FURIOUS: Fully unified risk-assessment with interactive operational user system for vessels ▪ 2024 INFORMS Annual Meeting (Contributed Session) Oct 2024 Spatio-Temporal Dynamics-Aware GNN Framework for Vessel Behavior Anomaly Detection ▪ 2025 Korea Data Mining Society (Best Paper Award) Aug 2025
REFERENCES	Prof. Hyunwoo Park , Associate Professor, Seoul National University ▪ Email: hyunwoopark@snu.ac.kr

[CV compiled on 2025-12-15]